Thm 10.3

Let $\Omega \in \mathbb{R}^n$ be a domain, let f be a continuous func.
g be a continuous func. on $\partial\Omega$
Assume $u, v \in C^2(\Omega) \cap C(\overline{\Omega})$ are ~~satisfy~~ sol. of

$$\begin{cases} -\Delta u(x) = f(x) \; ; \; x \in \Omega \\ u(x) = g(x) \; ; \; x \in \partial\Omega \end{cases}$$

$$\begin{cases} -\Delta v(x) = f(x) \; ; \; x \in \Omega \\ v(x) = g(x) \; ; \; x \in \partial\Omega \end{cases}$$

then $u = v$ in Ω

pf. let $w = u - v$

$$\Rightarrow \begin{cases} -\Delta(u-v)(x) = 0 \; ; \; x \in \Omega \\ (u-v)(x) = 0 \; ; \; x \in \partial\Omega \end{cases}$$

$$\Rightarrow \begin{cases} -\Delta w(x) = 0 \; ; \; x \in \Omega \\ w(x) = 0 \; ; \; x \in \partial\Omega \end{cases}$$

By weak maximum principle

$$\min_{x\in\partial\Omega} w(x) \le w(x) \le \max_{x\in\partial\Omega} w(x) \; ; \; \forall x \in \Omega$$

$$0 \le w(x) \le 0 \Rightarrow w(x) = 0 \; ; \; \forall x \in \Omega$$

$$\therefore \; u(x) - v(x) = 0.$$

$$\Leftrightarrow u(x) = v(x) \; ; \; \forall x \in \Omega \quad \boxtimes$$

Chapter 11 Green func.

Problem:

For open sets $\Omega \in \mathbb{R}^n$ and for given $f : \Omega \to \mathbb{R}^n$, $g : \partial\Omega \to \mathbb{R}$ consider
to ~~discri~~ derive the sol. formula of.

$$\begin{cases} -\Delta u(x) = f(x) \; ; \; x \in \Omega \\ u(x) = g(x) \; ; \; x \in \partial\Omega \end{cases} \quad (11.1).$$

For $x \in \Omega$ take $\varepsilon > 0$ satisfying $B_\varepsilon(x) \subset \Omega$
Let Φ be the fundamental sol of. Laplace eq. (def (9.2))

$$\int_{\Omega \setminus B_\varepsilon(x)} \left(u(y)\, \Phi(\overset{y-x}{\cancel{x-y}}) - \Phi(\overset{y-x}{\cancel{x-y}})\, u(y) \right) dy$$

$$\underset{\text{div thm.}}{=} \int_{\partial(\Omega \setminus B_\varepsilon(x))} u(y) \nabla \Phi(\overset{y-x}{\cancel{x-y}}) \cdot \nu(y) - \Phi(y-x) u(y) \cdot \nu(y)\, d\sigma(y)$$

$$- \int_{\Omega} \nabla u(y) \cdot \nabla \Phi(y-x)\, dy + \int_{\Omega} \nabla \Phi(y-x) \cdot \nabla u(y)\, dy$$

Using $\Delta \Phi(y-x) = 0 \; (\forall y \in \Omega \setminus B_\varepsilon(x))$.

$$- \int_{\Omega} \Phi(y-x) \Delta u(y)\, dy = \int_{\partial\Omega} \Big(u(y) \nabla_y \Phi(y-x)\, \nu(y)$$

$$- \Phi(y-x) \nabla_y \overset{u}{\cancel{\Phi}}(y-x) \cdot \nu(y) \Big) d\sigma(y)$$

$$+ \int_{\partial B_\varepsilon(x)} \Big(u(y) \nabla_y \Phi(y-x) \underbrace{\nu(y)}_{\frac{x-y}{\varepsilon}} - \Phi(y-x) \nabla_y u(y-x) \cdot \underbrace{\nu(y)}_{\frac{x-y}{\varepsilon}} \Big) d\sigma$$

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Since $\nabla \Phi(y-x) = \dfrac{1}{n\alpha(n)} \cdot \dfrac{\cancel{x}\ \ y-x}{\cancel{|x|^n}\ \ |y-x|^n}$ (from (9.2)) (Exercise).

$$\int_{\partial B_\varepsilon(x)} u(y)\, \nabla_y \Phi(y-x) \cdot \nu(y)\, d\sigma(y)$$

$$= \int_{\partial B_\varepsilon(x)} u(y) \cdot \frac{1}{n\alpha(n)} \cdot \frac{y-x}{|y-x|^n} \cdot \frac{y-x}{\varepsilon}\, d\sigma(y)$$

$$= \int_{\underbrace{\partial B_\varepsilon(x)}_{\{y\in\mathbb{R}^n;\ |y-x| = \varepsilon\}}} u(y) \cdot \frac{1}{n\alpha(n)}\, \underbrace{|y-x|^{2-n}}_{\varepsilon^{2-n}} \cdot \varepsilon^{-1}\, d\sigma(y)$$

$$= \frac{1}{\varepsilon^{n-1}\, n\alpha(n)} \underbrace{\int_{\cancel{\partial B(\varepsilon)}\ \partial B_\varepsilon(x)} d\sigma(y)}_{|\partial B_\varepsilon(x)|}$$

$\to u(\varepsilon) \quad (\varepsilon \downarrow 0)$

$$\left| \int_{\partial B_\varepsilon(x)} \Phi(y-x)\, \nabla_y u(y) \cdot \nu(y)\, d\sigma(y) \right|$$

$$\leq \int_{\partial B_\varepsilon(x)} |\Phi(y-x)| \cdot |\nabla_y u(y)| \cdot \underbrace{|\nu(y)|}_{=1}\, d\sigma(y)$$

$$\leq \sup_{y\in\Omega} |\nabla_y u(y)| \int_{\partial B_\varepsilon(x)} |\Phi(y-x)|\, d\sigma(y)$$

Note that

$$|\Phi(y-x)| = \begin{cases} \text{Const} \ ; \left|\log|y-x|\right| & ; n=2 \\ \text{Const} \ ; \dfrac{1}{|y-x|^{n-2}} & ; n \geq 3 \end{cases}$$

Thus.

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$$\int_{\partial B_\varepsilon(x)} |\Phi(y-x)|\, d\sigma(y)$$

$$\leq \begin{cases} \text{const.}\ \log\varepsilon \ \overbrace{|\partial B_\varepsilon(x)|}^{2\pi\varepsilon} & ; n=2 \\ \text{const.}\ \varepsilon^{2-n}\ \underbrace{|\partial B_\varepsilon(x)|}_{\text{const.}\,\varepsilon^{n-1}} & ; n\geq 3 \end{cases}$$

$\to 0 \quad (\varepsilon \downarrow 0)$.

Taking $\varepsilon \downarrow 0$.

$$\int_\Omega \Phi(y-x)\, \Delta u(y)\, dy$$

$$= \int_{\partial\Omega} \Big(u(y)\, \nabla\Phi(y-x) - \Phi(y-x)\, \nabla u(y) \Big)\, d\sigma(y) + u(x) \qquad (1\phi.3)$$

(for any $u \in C^2(\Omega) \cap C(\overline{\Omega})$).

To deal with $\Phi(y-x)\nabla u(y)$ on $\partial\Omega$ let ϕ^x be a sol of.

$$\begin{cases} -\Delta_y \phi^x(y) = 0 & ; y \in \Omega \\ \phi^x(y) = \phi(y-x) & ; y \in \partial\Omega \end{cases} \qquad (1.14)$$

Then.

$$-\int_\Omega \phi^x(y)\, \Delta u(y)\, dy$$

$$\underset{\text{div. thm}}{=} -\int_{\partial\Omega} \phi^x(y)\, \nabla u(y) \cdot \nu(y)\, d\sigma(y)$$

$$+ \int_\Omega \nabla_y \phi^x(y)\, \nabla u(y)\, dy$$

$$\underset{\text{div. thm}}{=} -\int_{\partial\Omega} \underbrace{\phi^x(y)}_{\Phi(y-x)}\, \nabla u(y) \cdot \nu(y)\, d\sigma(y)$$

$$+ \int_{\partial\Omega} u(y)\, \nabla_y \phi^x(y) \cdot \nu(y)\, d\sigma(y)$$

$$- \int_\Omega \underbrace{\Delta_y \phi^x(y)}_{=0} \cdot u(y)\, dy$$

$$= -\int_{\partial\Omega} \Phi(y-x)\,\nabla u(y)\cdot\nu(y)\,d\sigma(y)$$
$$+ \int_{\partial\Omega} u(y)\,\nabla_y \phi^{x}(y)\cdot\nu(y)\,d\sigma(y)$$

Plugging this relation into (11.3)

$$-\int_{\Omega}\Big(\phi(y-x)-\phi^{x}(y)\Big)\Delta u(y)\,dy = \int_{\partial\Omega}\Big(u(y)\quad \nabla\phi(y-x)-\nabla\phi^{x}(y)\Big)$$
$$\nu(y)\,d\sigma(y) + u(x)$$

Define $G(x,y) := \Phi(y-x) - \phi(\overset{y}{\cancel{x}})$, then

$$u(x) = -\int_{\Omega} G(x,y)\,\Delta u(y)\,dy - \int_{\partial\Omega} u(y)\,\nabla_y G(x,y)\cdot\nu(y)\,d\sigma(y)$$

Def. (Green func.)

For $x, y\in\Omega$ with $x\neq y$

$G(x,y) := \Phi(y-x) - \phi^{x}(y)$

is called the Green func. for (11.1), where

Φ is the fundamental sol of the Laplace eq.

and $\phi^{x}(y)$ is a sol of.

$$\begin{cases} -\Delta\phi^{x}(y) = 0, & y\in\Omega \\ \phi^{x}(y) = \Phi(y-x), & y\in\partial\Omega \end{cases}$$ ▨

Thm. 11.1: Let $u\in C^2(\Omega)\cap C^2(\overline{\Omega})$ be a sol of (11.1)

Then

$$u(x) = -\int_{\Omega} G(x,y) f(y)\,dy - \int_{\partial\Omega} g(y)\,\nabla_y G(x,y)\cdot\nu(y)\,d\sigma(y)$$

for any $x\in\Omega$ ▨

It is difficult to represent explicity the Green func.

Thm 11.2

For $x, y\in\Omega$ with $x\neq y$,

$G(x,y) = G(y,x)$ ▨ → ref lec. note

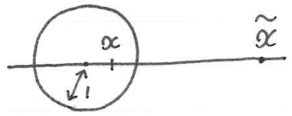
§ Green func on the ball

Let $\Omega = B_1(0) = \{y\in\mathbb{R}^n;\ |y|<1\}$

we want to derive the Green func. on Ω.

For $x\in\Omega = B_1(0)$.

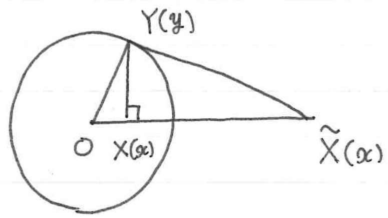
Let $\tilde{x} := \dfrac{x}{|x|^2}$



Consider

$$\begin{cases} -\Delta\phi^{x}(y) = 0 & ;\ y\in\Omega \\ \phi^{x}(y) = \Phi(y-x) & ;\ y\in\partial\Omega \end{cases} \qquad (11.6)$$

Fix $y\in\partial\Omega = \partial B_1(0)$



$\triangle OYX \backsim \triangle O\tilde{X}Y$

∵ $\triangle OYX$ と $\triangle O\tilde{X}Y$ について

- $\angle XOY = \angle YO\tilde{X}$
- $OY : OX = 1 : |x|$
- $O\tilde{X} : OY = |\tilde{x}| : 1 = |x| \ (|\tilde{x}| : |x| = 1 : |x|)$

∴ $OY : OX = O\tilde{X} : OY$

従って 1つの角とそれを挟む辺の比がそれぞれ等しいので

$\triangle OYX \backsim \triangle O\tilde{X}Y$ ▨

therefore.

$$\tilde{X}Y : YO = YX : XO$$

$$\Leftrightarrow |\tilde{x}-y| : |y| = |x-y| : |x|$$

$$\Leftrightarrow |x|\cdot|\tilde{x}-y| = |y|\cdot|x-y|.$$

$$\therefore\ |x|\cdot|\tilde{x}-y| = |x-y| \quad (\because y\in\partial B_1(0))$$

Thus $\Phi(y-x) = \Phi(|x|(\tilde{x}-y)) \quad (\because \forall y\in\partial\Omega=\partial B_1(0))$

$$= \Phi(|x|(y-\tilde{x})).$$

Since

$$\Delta_y \Phi(|x|(y-\tilde{x})) = 0 \quad \forall y\in B_1(0)$$

$\phi^x(y) = \Phi(|x|(y-\tilde{x}))$ is a sol of (11.6)

$$\Phi(|x|(y-\tilde{x})) = \begin{cases} -\dfrac{1}{2\pi}\log\left(|x|\cdot|y-\tilde{x}|\right) & ;\ n=2 \\ \dfrac{1}{n(n-2)\alpha(n)}\dfrac{1}{(|x|\cdot|y-\tilde{x}|)^{n-2}} & ;\ n\geq 3 \end{cases}$$

Thm 11.3

The Green func. of (11.1) with $\Omega = B_1(0)$ can be represented as

$$G(x,y) = \begin{cases} -\dfrac{1}{2\pi}\left(\log|x-y| + \log\left(|x|\cdot\left|y-\dfrac{x}{|x|^2}\right|\right)\right) & ;\ n=2 \\ \dfrac{1}{n(n-2)\alpha(n)}\left(\dfrac{1}{|x-y|^{n-2}} + \dfrac{1}{|x|\cdot\left|y-\dfrac{x}{|x|^2}\right|^{n-2}}\right) & n\geq 3 \end{cases}$$

Let $g:\partial B_1(0)\to\mathbb{R}$ be a given continuos func. Consider

$$\begin{cases} -\Delta u(x) = 0, & x\in B_1(0) \\ u(x) = g(x); & x\in\partial B_1(0) \end{cases} \qquad (11.8)$$

By Thm 11.1 or 11.2

$$u(y) = -\int_{\partial B_1(0)} g(y)\,\nabla_y G(x,y)\cdot\nu(y)\,d\sigma(y).$$

By direct computation of $\nabla_y G(x,y)$, we have

$$u(y) = \frac{1-|x|^2}{n\alpha(n)}\int_{\partial B_1(0)}\frac{g(y)}{|x-y|^n}\,d\sigma(y)$$

(Poisson Integral)

(ref. Gilbarg - Trudinger. (1982) or reprint ver.).